



Modelling Financial Markets

Seminar 13: Machine Learning in Finance — Hands-on



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Acronyms used in this material (1/2)

- ▣ ADF: Augmented Dickey–Fuller (test)
- ▣ AI: Artificial Intelligence
- ▣ AUC: Area Under the (ROC) Curve
- ▣ CI: Confidence Interval
- ▣ CSCV: Combinatorially Symmetric Cross-Validation
- ▣ CV: Cross-Validation
- ▣ DM: Diebold–Mariano (test)
- ▣ DSR: Deflated Sharpe Ratio
- ▣ EWMA: Exponentially Weighted Moving Average
- ▣ FFD: Fixed-width window Fractional Differentiation
- ▣ FN: False Negative
- ▣ GB: Gradient Boosting
- ▣ HAC: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- ▣ HP: Hodrick–Prescott (filter)
- ▣ LASSO: Least Absolute Shrinkage and Selection Operator
- ▣ MA: Moving Average
- ▣ MA20: 20-day Moving Average
- ▣ MDI: Mean Decrease Impurity
- ▣ ML: Machine Learning
- ▣ PBO: Probability of Backtest Overfitting
- ▣ PSR: Probabilistic Sharpe Ratio
- ▣ RF: Random Forest
- ▣ ROC: Receiver Operating Characteristic (curve)
- ▣ RSI: Relative Strength Index
- ▣ SE: Standard Error
- ▣ SR: Sharpe Ratio

Acronyms used in this material (2/2)

- **TN: True Negative**
- **TP: True Positive**

Outline

Objectives and Setup

Part A: Pen-and-Paper Problems

Part B: Python Problems on Crypto Data

Part C: Open Problem and AI Critique

Assignment and Further Study

Key Definitions (1/3): Data, Labels, Validation

Features and labels

- **FFD**: $\tilde{X}_t = \sum_{k=0}^K w_k X_{t-k}$, weights truncated when $|w_k| < \tau$
- **ADF (Dickey & Fuller (1979))**: H_0 : unit root; reject (stationary) if the statistic is below the 5% critical value
- **EWMA**: standard deviation of daily log returns with exponentially decaying weights (span 50)
- **Triple barrier**: label $+1/-1$ if the upper/lower barrier (pt , sl multiples of $\sigma_{t_0} \sqrt{h}$) is hit first; else the sign at $t_0 + h$

Validation

- **K-fold CV**: K blocks; fit on $K - 1$, test on 1, repeat
- **Leakage**: training data contain information that is also in the test data
- **Purging / embargo**: drop training observations whose label span overlaps the test block, and a few just after it

Key Definitions (2/3): Models and Classification Metrics

Models and signals

- **Logit-L1**: logistic regression with an L_1 (LASSO) penalty
- **RF**: average of decorrelated trees; **GB**: small trees added sequentially
- **MA crossover**: long when the fast MA is above the slow one

Classification metrics

- Accuracy = $\frac{TP+TN}{n}$; precision = $\frac{TP}{TP+FP}$; recall = $\frac{TP}{TP+FN}$;
 $F_1 = \frac{2PR}{P+R}$ (TP, FP, TN, FN: true/false positives/negatives)
- **ROC AUC**: $P(\hat{s}_+ > \hat{s}_-)$ for a random positive and a random negative; 0.5 = no skill
- **Log loss**: $\ell_t = -[y_t \ln \hat{p}_t + (1 - y_t) \ln(1 - \hat{p}_t)]$, the loss of a probability forecast

Key Definitions (3/3): Inference and Backtests

Uncertainty

- ▣ **95% CI**: estimate ± 1.96 SE
- ▣ **Block bootstrap**: resample blocks of L consecutive observations to keep serial dependence (Künsch (1989))
- ▣ **HAC**: Newey–West variance (Newey & West (1987)), valid under serial correlation
- ▣ **DM**: Diebold–Mariano test of equal predictive accuracy: t -test on the loss differential $d_t = \ell_t^A - \ell_t^B$ with HAC SE (Diebold & Mariano (1995))

Backtests

- ▣ **SR**: $\bar{r}/\hat{\sigma}$; annualised $\times \sqrt{365}$ for crypto
- ▣ **PSR / DSR** (Bailey & López de Prado (2012); Bailey & López de Prado (2014)): use skewness γ_3 , kurtosis γ_4 (Normal distribution: 3)
- ▣ N_{eff} : effective number of independent trials
- ▣ **PBO**: estimated by CSCV (Bailey, Borwein, López de Prado & Zhu (2017))

How This Seminar Works

- ▣ Two kinds of problems, in every part
 - ▶ **[Solved]**: worked models, with full solutions in this deck and in the notebook
 - ▶ **[Proposed]**: you solve them; each follows an analogous solved problem
- ▣ Part A: pen and paper (derivations, small numerical cases)
- ▣ Part B: Python on Bitcoin, verification *and* inference: every number with its uncertainty
 - ▶ each exercise ends with interpretation questions: what would change the conclusion?
- ▣ Part C: an open question that can seed your team project (C1)
 - ▶ and an AI answer to critique (C2): find the leaks, correct them, log them in `AI_ERRORS.md`
- ▣ Rules: no look-ahead, purged validation, a naive benchmark, and a log of every trial ([Harvey, Liu & Zhu \(2016\)](#))

Setup: Python Environment and Data

Environment

- Google Colab (recommended) or local Python ≥ 3.10
- `numpy`, `pandas`, `scipy`, `statsmodels`, `scikit-learn`, `shap`
-  **Open in Colab:**
`chapter13_seminar_notebook.ipynb`

```
import pandas as pd
url = 'https://raw.githubusercontent.com/danpele/MFM/main/data/market/BTC-USD.CC.csv'
btc = pd.read_csv(url, parse_dates=['date']).set_index('date')['close'].loc['2014-09-17':]
```

Data

- **Bitcoin in dollars (BTC-USD)**, daily, 2014–2026 (4 385 obs.)
- **Ether (ETH-USD)**, daily, 2015–2026 (4 061 obs.)
- Files in the course repository: `data/market`

A1 [Solved]: Fractional Differentiation Weights

Task

- ▣ $\tilde{X}_t = \sum_{k \geq 0} w_k X_{t-k}$, $w_0 = 1$, $w_k = -w_{k-1} \frac{d-k+1}{k}$
(Hosking (1981); López de Prado (2018))
- ▣ Compute $w_0, \dots, w_4$ for $d = 0.5$
- ▣ Check $d = 0$ and $d = 1$: what do they give?
- ▣ Why does $0 < d < 1$ keep memory?

Solution

- ▣ $w = (1; -0.5; -0.125; -0.0625; -0.0391)$
- ▣ $d = 0$: $w = (1, 0, \dots) \Rightarrow$ prices; $d = 1$:
 $w = (1, -1, 0, \dots) \Rightarrow$ returns
- ▣ For $0 < d < 1$ all $w_k < 0$ ($k \geq 1$) decay hyperbolically: $\tilde{X}_t$ is a long weighted sum of past prices
- ▣ Integer differencing ($d = 1$) throws this memory away

A2 [Proposed]: Weights and Window Length

Task

- ▣ Compute $w_0, \dots, w_5$ for $d = 0.3$
- ▣ With threshold $\tau = 0.01$, how many weights does FFD keep for $d = 0.3$ and for $d = 0.5$?
- ▣ Explain why a smaller d needs a longer window and loses more observations at the start of the sample

Solution

- ▣ Solved by you; discussed in class. Model: the solved problem just before

A3 [Solved]: Triple-Barrier Label

Task

- $P_{t_0} = 100$, $\sigma_{t_0} = 1\%$, $h = 5$, $pt = sl = 1$: barriers
 $P_{t_0} e^{\pm \sigma_{t_0} \sqrt{h}}$ (López de Prado (2018))
- Label the two paths (days 0–5)

Day	0	1	2	3	4	5
1	100	101.0	100.4	102.5	101.2	99.8
2	100	99.2	98.6	99.5	98.9	99.1

Solution

- Width $0.01\sqrt{5} = 0.02236$: upper 102.26, lower 97.79
- Path 1: day 3, $102.5 \geq 102.26 \Rightarrow +1$ (the later drop is irrelevant)
- Path 2: no horizontal barrier; $\ln(99.1/100) = -0.90\%$ at day 5 $\Rightarrow -1$
- A fixed 5-day label would give path 1 the label -1 , although a take-profit order gained

A4 [Proposed]: Asymmetric Barriers

Task

- ▣ $P_{t_0} = 50$, $\sigma_{t_0} = 2\%$, $h = 4$, $pt = 2$, $sl = 1$
- ▣ Compute both horizontal barriers and label the paths: (a) 50, 51.5, 53.0, 49.5, 47.9; (b) 50, 52.0, 54.5, 53.0, 51.0
- ▣ With $pt = 2$, $sl = 1$ and a driftless price, which label is more frequent among horizontal touches? Why is this a risk-management choice, not a forecast?

Solution

- ▣ Solved by you; discussed in class. Model: the solved problem just before

A5 [Solved]: PSR and DSR

Task

- $\widehat{SR}_{ann} = 1.2$, $T = 756$ daily obs., $\hat{\gamma}_3 = -0.4$, $\hat{\gamma}_4 = 5$
- $PSR(SR^*) = \Phi\left(\frac{(\widehat{SR} - SR^*)\sqrt{T-1}}{\sqrt{1 - \hat{\gamma}_3\widehat{SR} + \frac{\hat{\gamma}_4 - 1}{4}\widehat{SR}^2}}\right)$ (Bailey & López de Prado (2012))
- The strategy is the best of $N = 100$ independent trials:
 $SR_0 = \sqrt{V}\left[(1 - \gamma)\Phi^{-1}\left(1 - \frac{1}{N}\right) + \gamma\Phi^{-1}\left(1 - \frac{1}{N_e}\right)\right]$,
 $\gamma = 0.5772$, $V = 1/T$; $DSR = PSR(SR_0)$ (Bailey & López de Prado (2014))

Solution

- Daily $\widehat{SR} = 1.2/\sqrt{252} = 0.07559$; denominator 1.01782
- $PSR(0) = \Phi(2.041) = 0.979$
- $\sqrt{V} = 0.03637$; $SR_0 = 0.0920$ (annualised 1.46)
- $DSR = \Phi(-0.444) = 0.329$: luck alone gives a best SR of $1.46 > 1.2$; do not allocate

A6 [Proposed]: PSR and DSR, Shorter Sample

Task

- ▣ $\widehat{SR}_{ann} = 1.5$ on 2 years of daily data ($T = 504$), $\hat{\gamma}_3 = -0.2$, $\hat{\gamma}_4 = 6$
- ▣ Compute $PSR(0)$, then SR_0 and DSR if the strategy is the best of $N = 20$ independent trials ($V = 1/T$)
- ▣ Compare with A5: a higher Sharpe ratio and fewer trials, yet is the evidence stronger? What drives the difference?

Solution

- ▣ Solved by you; discussed in class. Model: the solved problem just before



A7 [Solved]: Purging and Embargo

Task

- $n = 20$, label window $[i, t_1(i)]$, $t_1(i) = \min(i + 3, 19)$; 5 unshuffled folds, embargo 1 observation (López de Prado (2018))
- (a) Why does shuffled K-Fold overstate accuracy?
- (b) Test fold $\{8, \dots, 11\}$: which observations are purged / embargoed?

Solution

- (a) Neighbours share 3 of 4 label days: shuffling puts near-duplicates in train and test (lecture: 69.4% shuffled vs 49.7% purged on pure noise)
- (b) Purged before: $\{5, 6, 7\}$ ($t_1 \geq 8$); removed after: $\{12, 13, 14\}$ (12 = embargo; 13–14 start before the last test label, day 14, is known)
- Training set $\{0, \dots, 4\} \cup \{15, \dots, 19\}$

A8 [Proposed]: Purging with Shorter Labels

Task

- ▣ $n = 20$, $t_1(i) = \min(i + 2, 19)$, 5 unshuffled folds, embargo 1 observation
- ▣ For the test fold $\{12, \dots, 15\}$: list the purged, the embargoed and the training observations
- ▣ How does the number of removed observations scale with the label horizon? What does this cost when $h = 20$ and folds are short?

Solution

- ▣ Solved by you; discussed in class. Model: the solved problem just before

A9 [Solved]: AUC Step by Step, with a Standard Error

Task

- Scores of 3 positives: 0.80, 0.55, 0.40; of 3 negatives: 0.60, 0.35, 0.20
- (a) AUC = share of (positive, negative) pairs ranked correctly (Mann–Whitney $U/(n_1 n_0)$)
- (b) Hanley–McNeil SE (Hanley & McNeil (1982)):

$$SE^2 = \frac{A(1-A) + (n_1 - 1)(Q_1 - A^2) + (n_0 - 1)(Q_2 - A^2)}{n_1 n_0}, \quad Q_1 = \frac{A}{2-A},$$

$$Q_2 = \frac{2A^2}{1+A}$$
- (c) Is the AUC significantly above 0.5?

Solution

- (a) Correct pairs: 0.80 beats 3, 0.55 beats 2, 0.40 beats 2 $\Rightarrow U = 7, A = 7/9 = 0.778$
- (b) $Q_1 = 0.636, Q_2 = 0.681 \Rightarrow SE = 0.207$
- (c) $z = (0.778 - 0.5)/0.207 = 1.34$: not significant; 6 observations cannot tell skill from luck
- The formula assumes independent observations: in B7 we replace it by a block bootstrap

A10 [Proposed]: AUC Step by Step

Task

- ▣ Positives: 0.70, 0.62, 0.48, 0.45; negatives: 0.66, 0.50, 0.40, 0.30
- ▣ Compute the AUC, its Hanley–McNeil SE and the z-statistic against 0.5
- ▣ How many observations would you need, at this AUC, for $z \approx 2$? Why is that number too optimistic for overlapping 5-day labels?

Solution

- ▣ Solved by you; discussed in class. Model: the solved problem just before

B1 [Solved]: Minimum d^* for Bitcoin

Task

- FFD of $\ln P_t^{BTC}$ ($\tau = 10^{-4}$) for $d \in \{0, 0.05, \dots, 1\}$; ADF (constant, 1 lag) and $\text{Corr}(\tilde{x}_t, x_t)$
- Smallest d^* with ADF below the 5% critical value

Interpretation

- What would change d^* : the deterministic terms of the ADF (constant vs trend), the sample start, the lag length?

Solution (BTC 2014–2026)

d	0	0.05	0.10	0.15	0.20	0.25
ADF	-1.21	-2.32	-2.24	-2.39	-2.85	-3.38
Corr	1.000	0.999	0.995	0.985	0.972	0.955

- 5% critical value $-2.86 \Rightarrow d^* = 0.25$, correlation 0.955 with the log price
- The ADF path is not monotone in d (-2.32 then -2.24): read the whole curve, not one number



B2 [Proposed]: Minimum d^* for Ether — a Trap

Task

- Repeat B1 on $\ln P_t^{ETH}$ (2015–2026). What d^* do you get?
- Redo the $d = 0$ test with a constant *and a trend*, and redo B1 on the sample starting in 2018
- Interpretation: which d^* would you use in a model, and why? What does this say about unit-root tests on assets that grew 1 000-fold?

Solution

- Solved by you; discussed in class. Model: the solved problem just before

B3 [Solved]: Triple-Barrier Labels on Bitcoin

Task

- EWMA volatility (span 50), $h = 10$; label every day; $pt = sl \in \{1, 2\}$
- Shares of upper / lower / vertical exits and of +1 labels

Interpretation

- Why are 56% of labels +1, and what is then the benchmark accuracy of a classifier?

Solution

	Upper	Lower	Vertical	+1
$pt = sl = 1$	22.7%	18.2%	59.1%	55.8%
$pt = sl = 2$	7.2%	4.9%	87.9%	55.7%

- Wider barriers push most exits to the vertical barrier: labels become fixed-horizon labels
- BTC rose over the sample: +1 dominates, so “always up” is the benchmark to beat

B4 [Proposed]: Asymmetric Barriers on Bitcoin

Task

- ▣ Repeat B3 with $(pt, sl) = (2, 1)$ and $(1, 2)$; report exit shares, +1 share and average holding period
- ▣ Check your answer to A4 on real data
- ▣ Interpretation: which setting would a trend follower choose? How would the choice change the class balance a classifier sees?

Solution

- ▣ Solved by you; discussed in class. Model: the solved problem just before

B5 [Solved]: Purged K-Fold vs Shuffled K-Fold

Task

- Write `PurgedKfold(n_splits, t1, pct_embargo)` with a `scikit-learn split(X)`
- Leakage dataset: random walk, label = sign of the 20-day forward return, 5 features = 20-day means of independent noise
- Random forest ([Breiman \(2001\)](#)): shuffled K-Fold vs Purged K-Fold

Interpretation

- Would the gap shrink with non-persistent features? With 1-day labels?

Solution (seed 42)

- Shuffled K-Fold: **69.4%**; Purged + 1% embargo: **49.7%**; truth: 50%
- Both ingredients are needed: overlapping labels *and* persistent features; remove either and the leak disappears
- Check: on the A7 example the class drops {5, 6, 7, 12, 13, 14}

B6 [Proposed]: Find the Leak

Task

- ▣ The notebook adds a 13th feature, `ma11`, to the B7 feature set; with Purged K-Fold the random forest now reaches a spectacular accuracy on BTC 5-day direction
- ▣ Purging did not stop it. Find the leak (hint: read how `ma11` is built and rank features by MDI), fix it and report the honest result
- ▣ Interpretation: why can purging not catch this kind of leak? Give a second example of look-ahead in a feature pipeline

Solution

- ▣ Solved by you; discussed in class. Model: the solved problem just before

B7 [Solved]: Model Comparison and a CI for the AUC

Task

- Target: sign of the 5-day forward log return; 12 features (momentum, volatility, MA distances, RSI)
- Logit-L1 (Tibshirani (1996)), RF, GB (Friedman (2001)); Purged 5-Fold; MDI/MDA/SHAP (Lundberg & Lee (2017))
- Pooled out-of-fold AUC of the RF: DeLong SE (DeLong, DeLong & Clarke-Pearson (1988)) vs circular block bootstrap ($L = 20$, 1 000 replications)

Interpretation

- Why is the bootstrap interval wider? Which block length would you defend?

Solution (4 181 obs., 54.8% “up”)

Fold mean	Logit-L1	RF	GB
Accuracy	54.8%	54.2%	52.2%
ROC AUC	0.492	0.516	0.502

- RF pooled AUC 0.512: DeLong SE 0.009, 95% interval [0.495; 0.530], $p = 0.17$ against 0.5
- Block bootstrap: SE 0.015 (1.7 times larger), 95% interval [0.484; 0.543]; 21% of replications ≤ 0.5
- DeLong assumes independent observations; overlapping labels and volatility clusters violate it. No model beats “always up”

B8 [Proposed]: Diebold–Mariano on Probability Forecasts

Task

- Use the pooled out-of-fold probabilities of B7 and a constant forecast (the training share of “up” days in each fold)
- Loss differential $d_t = \ell_t^A - \ell_t^B$ (log loss); DM statistic $= \bar{d}/SE_{HAC}(\bar{d})$ with Newey–West lags 4 and 20; compare with the naive i.i.d. t
- Pairs: RF vs Logit-L1, RF vs constant, GB vs constant
- Interpretation: can a model with $AUC > 0.5$ lose to a constant in log loss? What would change the conclusion?

Solution

- Solved by you; discussed in class. Model: the solved problem just before

B9 [Solved]: Meta-Labelling

Task

- Primary: side of the MA20/MA50 crossover; meta-label = 1 if the primary trade is profitable at the triple-barrier exit ($pt = sl = 1$, $h = 10$) (López de Prado (2018))
- Secondary: RF on features + side; train on the first 60%, 10-day gap, test on the rest

Interpretation

- Precision up, F1 down: which metric matters for a trader who pays costs per trade?

Solution (2022-02 to 2026-09, 1 660 signals)

	Prec.	Recall	F1
Primary only	0.475	1.000	0.644
+ meta-model	0.517	0.541	0.529

- The meta-model keeps about half of the trades; precision rises by 4 points, recall halves
- Per trade, fewer and better trades can win after costs even when F1 falls

B10 [Solved]: Grid Search, DSR and the Effective Number of Trials

Task

- Long/cash MA crossover on BTC (2015–2026): fast $\in \{5, \dots, 55\}$, slow $\in \{20, \dots, 240\} \Rightarrow N = 120$ trials; best by Sharpe; PSR and DSR (Bailey & López de Prado (2014))
- Effective N from the 120×120 correlation matrix of trial returns: eigenvalues λ_i , participation ratio $(\sum \lambda_i)^2 / \sum \lambda_i^2$, and $\hat{N} = \bar{\rho} + (1 - \bar{\rho})N$ (Bailey & López de Prado (2014))

Interpretation

- Which N and which $V[\widehat{SR}]$ would you report to an investment committee?

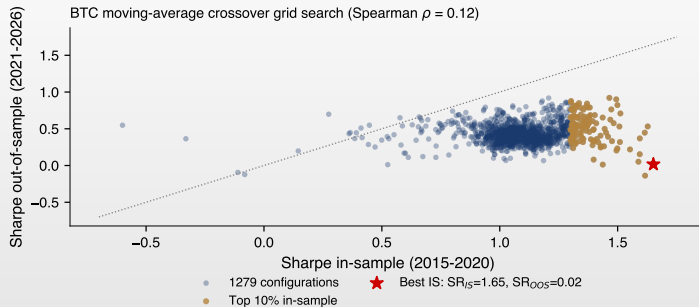
Solution

- Best (5, 40), $SR_{ann} = 1.17$ (buy & hold 0.71); $T = 4\,279$, skew 0.11, kurtosis 13.0
- $\lambda_1 = 105.2$ explains 87.7% of the variance; participation ratio 1.3; 4 eigenvalues reach 95%; $\bar{\rho} = 0.873 \Rightarrow \hat{N} = 16.1$

DSR, $V = 1/T$	$N = 120$	$\hat{N} = 16.1$	$N = 4$
SR_0 (ann.)	0.76	0.53	0.31
DSR	0.922	0.986	0.998

- Counting 120 correlated trials as independent over-deflates; the real question is “above buy & hold?”

B10: Why Out-of-Sample Evidence Still Matters



- Lecture result: the best in-sample BTC crossover (SR 1.65 in 2015–2020) earned SR 0.02 in 2021–2026

 [MFM_ch13_backtest_overfitting](#)

B11 [Proposed]: Probability of Backtest Overfitting (CSCV)

Task

- Matrix of daily returns of the 120 B10 trials, split into $S = 16$ contiguous blocks; for each of the $\binom{16}{8} = 12\,870$ ways to choose 8 blocks as in-sample (the rest out-of-sample)
 - pick the best in-sample Sharpe; compute its relative rank ω out of sample and $\lambda = \ln \frac{\omega}{1-\omega}$
- $PBO = P(\lambda \leq 0)$ (Bailey, Borwein, López de Prado & Zhu (2017)); also report the median out-of-sample Sharpe of the in-sample winner
- Interpretation: is the grid overfit? How would S , or adding 1 000 random strategies to the grid, change PBO?

Solution

- Solved by you; discussed in class. Model: the solved problem just before

C1 [Proposed]: Does a Foundation Model Beat Boosting?

The question

- Does a time-series foundation model (Chapter 14), used zero-shot, beat gradient boosting for BTC 5-day direction once purged validation, a DM test and the DSR are applied?
- No single method: propose one and defend it

Deliverable

- A Quantlet folder in the team repository (code, Metainfo, one chart) and one slide with the conclusion and its main limitation

What the answer must contain

- A probability of “up” from the model’s forecast quantiles, evaluated only on dates after its pre-training data (otherwise the test leaks)
- The B7/B8 baselines: log loss of gradient boosting and of the constant forecast
- A DM test on log loss with HAC errors, and the DSR over *all* configurations you tried

C2 [Proposed]: Critique an AI Answer

- A student asks an AI assistant: “Predict tomorrow’s direction of Bitcoin from past returns and evaluate the model out of sample with 5-fold cross-validation”
- The assistant answers with this code (btc: daily closes, imports omitted):

```
r = np.log(btc).diff()
X = pd.DataFrame({'r1': r, 'r5': r.rolling(5).sum(), 'vol20': r.rolling(20).std(),
                  'trend10': r.rolling(10, center=True).mean()})
data = pd.concat([X, r.shift(-1).rename('fwd')], axis=1).dropna()
y = (data['fwd'] > 0).astype(int) # 1 if tomorrow is up
Xs = StandardScaler().fit_transform(data[X.columns])
cv = KFold(n_splits=5, shuffle=True, random_state=0)
model = LogisticRegression(penalty='l1', C=0.1, solver='liblinear')
auc = cross_val_score(model, Xs, y, cv=cv, scoring='roc_auc')
print(round(auc.mean(), 3))
```

- “Output: 0.697. An out-of-sample AUC of 0.70 on 4361 days is a strong, tradable signal; cross-validation is out of sample, so there is no overfitting”

C2 [Proposed]: Critique an AI Answer, Task

Task

- The answer contains **three** leaks of test information into training; everything else is correct
- For each error
 - ▶ say what is wrong and why
 - ▶ say how you would detect it: a check in the code, a result from B5–B7, a rule of the lecture
 - ▶ give the corrected code
- Rerun the corrected code; report the AUC
- Log the three errors as in `AI_ERRORS.md`: what was asked, what the AI answered, what was wrong, how you found out, the correct version

Interpretation

- Which days enter `trend10` on day t ? Which of them is the target?
- Is an AUC of 0.70 for next-day Bitcoin direction plausible, given B6 and B7?
- Why is “cross-validation is out of sample” not enough for time series?

Take-Home Assignment (Team Project)

- Choose one asset (equity index, stock, crypto, exchange rate or commodity) and deliver
 - ▶ features without look-ahead, including one FFD feature with a justified d^* and its sensitivity (B2)
 - ▶ triple-barrier labels with a justified (pt, sl, h)
 - ▶ at least three models with Purged K-Fold, a DM test against a constant forecast and a block-bootstrap CI for the AUC
 - ▶ a log of every trial, the effective N , the DSR and the PBO of the final strategy
- GitHub repository with notebooks and Quantlet Metainfo.txt; a 10-minute presentation
- Log every AI error you catch in AI_ERRORS.md, as in C2: what was asked, what was wrong, how you found out, the correct version
- A negative result, honestly documented, is a good result

Further Study on Quantinar

-  [Random Forests](#) — bagging, out-of-bag error, variable importance
-  [Shapley Values](#) — model-agnostic local explanations
-  [Machine Learning in Financial Risk](#) — ML estimation of risk measures
-  [Introduction to Reinforcement Learning](#) — Q-learning, sequential decisions

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